Video-Based People Counting

YOLO11, custom ByteTrack, and directional line counting



Research question: Do fine-tuning and detector fusion improve directional counting across crowded and changing views?

Dataset and evaluation protocol

Detector development

MOT17-04 and MOT17-05

Train: first 80%
1,510 images, 39,144 boxes

Validation: last 20%
377 images, 11,397 boxes

Comparison: 2,454 frames

MOT17-02, MOT20-01, MOT17-09, MOT17-11



MOT17-09: a different camera view

Comparison cameras were excluded from training, but used for model selection. All reported scores are preliminary post-selection comparisons.

Methodology

Three stages turn video into directional counts

01

DETECT

YOLO11 person boxes

COCO pretrained or fine-tuned

02

TRACK

Custom ByteTrack

Kalman prediction + two-stage IoU matching

03

COUNT

Line crossing events

Bottom-center point, hysteresis, cooldown

Second approach

Fuse the 3-epoch and COCO YOLO11s boxes before the same tracker and counter.

Results across five approaches

Unweighted mean across four comparison sequences

Approach	CA ↑	Curve MAE ↓	MOTA ↑	IDF1 ↑
COCO YOLO11n	0.5501	2.5913	**0.4361**	0.4446
COCO YOLO11s	0.6222	1.9747	0.4352	**0.4671**
YOLO11s, 3 epochs	0.6548	**1.7021**	0.4113	0.4324
YOLO11s, 20 epochs	0.5794	2.1542	0.4080	0.4277
Two-detector ensemble	**0.6654**	1.7858	0.3989	0.4261

The ensemble gains 0.0106 CA over the 3-epoch model, but has worse temporal count error and tracking metrics.

Preliminary post-selection comparison

Scene-level differences

Fusion helps crowding and hurts the two camera-view scenes

Sequence	3-epoch CA	Ensemble CA	Change
MOT17-02 occlusion	0.5238	0.5714	+0.0476
MOT20-01 dense crowd	**0.3462**	**0.5385**	**+0.1923**
MOT17-09 low-angle view	0.9231	0.7692	−0.1539
MOT17-11 moving view	0.8261	0.7826	−0.0435

MOT20-01

The 20-epoch model reaches CA 0.6154 here, but only 0.5794 mean CA across the four scenes.

Runtime on 2,454 frames

125.42 s single
164.62 s ensemble

Representative counting failures

Three event discrepancies verified against ground-truth counting events

Scene	Observed error	Saved diagnostic
MOT20-01, GT frame 70	IN missing in both approaches	Best ensemble track overlap: IoU 0.1135. Dense overlap may disrupt detection or association.
MOT17-09, predicted frame 252	Ensemble emits an extra OUT	Track 58 changes its best-matching person near the line. Apparent identity transfer.
MOT17-11, GT frame 146	IN missing in both approaches	Best ensemble track overlap: IoU 0.1978 in a moving view.

These observations identify the count errors. The saved outputs do not prove a unique detector or tracker cause.

Conclusions and limitations

3-epoch YOLO11s

Practical default: CA 0.6548 and the lowest Curve MAE, 1.7021.

Two-detector ensemble

Highest observed mean CA, 0.6654, but worse MOTA, IDF1, Curve MAE, and about 31% longer runtime.

What the study establishes

The implemented pipeline and trained checkpoint work on the four studied scenes.

What remains to test

The comparison scenes informed model choice. Transfer to new cameras needs an untouched evaluation.

Repository: github.com/NikolayTaran/cv_people_counting